# Coasters and pendulums

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use height differences to predict speed in ideal motion.

**SUCCESS CRITERIA - I can...**

1. Identify the relevant vertical height change.
2. Conserve energy only when the stated facts allow it.
3. Separate path length from gravitational energy change.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “pendulum” and “potential energy” to explain your answer. Sentence starters are optional.

**pendulum**: an object that swings about a fixed support

**potential energy**: energy associated with the configuration of interacting objects

**kinetic energy**: energy associated with motion

## 01 Warm-up: bring back an earlier idea

With GMm=**105 J·m**, move an object from r=**3** to r=**7 m**. **Find work** by gravity using U=-GMm/r.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** mg Δh = ½mv²; v = √(**2 g** Δh).
Point mass released from rest, no losses or rotation; Δh is the positive vertical drop.

**Worked reasoning:** A pendulum released from rest **0.8 m** above its lowest point reaches speed √(**2** gh)=**4 m/s** there with g=**10** and negligible losses.

**Your turn:** Sketch the path and energy bars at the release, bottom and opposite turning point.

**Sentence starter:** In my work, I used ... to show ... . I labeled it ... .

**Initial** **Transfers** **Final**

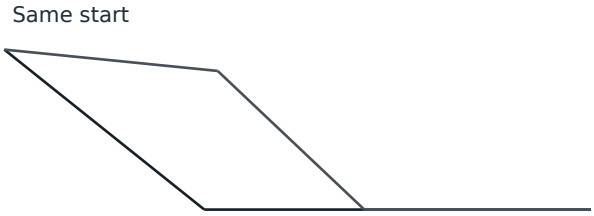
Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

Two frictionless tracks start and end at the same heights but take different paths. **Compare final speeds** from rest.

**Sentence starter:** Both ... . They differ in ... because ... .




Mark the vertical height change.

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A coaster enters height **8 m** at speed **6 m/s**.
1. **Find speed** at height **3 m** with no losses.
2. Use g=**10**.
3. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

**Sentence starter:** Both ... . They differ in ... because ... .

**Initial** **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Explain your reasoning

A classmate uses total ramp length as h in mgh. **Explain how** to **name the correct measurement**.

**Sentence starter:** For this answer, I treated … as true. I found … . This supports … because … .

**Marzano check - circle:** 1 2 3

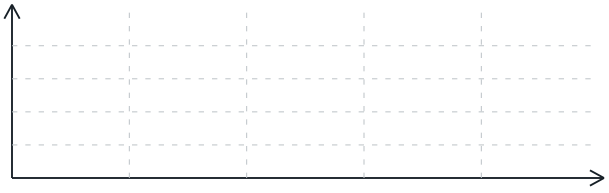
## 06 Review: connect an earlier idea

A positive triangular pulse peaks at **60 N** and lasts **0.4 s**. It acts on a **4 kg** cart initially at **-3 m/s**. **Find impulse** and final velocity.
**Show the signed impulse** on the force–time graph.

**Word help:** impulse = net force acting over time; change in momentum

**Sentence starter:** I found … [unit]. This value means … in this situation.

Force (N)



Time (s)

**Marzano check - circle:** 1 2 3

## 07 Exit check

A cart starts from rest at a height of **5 m**. Can it reach a height of **6 m** without energy from outside? **Explain why**. **Use the energy** diagram to **show the quantities you compare**. **Label units** and unknowns.

**Word help:** justify = **support your answer** with a reason or evidence

**Sentence starter:** Both … . They differ in … because … .

**Initial** **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

A frictionless coaster starts from rest at height **8 m** above the bottom of a vertical loop of radius **2 m**.
1. Use g=**10**.
2. **Find speed** and normal force per unit mass at the top; **check contact**.
3. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

**Sentence starter:** Both … . They differ in … because … .

**Initial** **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** 1 2 3